

A rank collapse in a sedenion Module-LWE proposal, and why a uniform matrix does not repair it

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October 2026

Abstract

We report a negative result on SLWE, a Module-LWE variant over the sixteen-dimensional sedenion algebra proposed in the author’s own research programme. Its public matrix is built from orbits of an order-7 coordinate permutation (a “Singer cycle”). First, the flattened public matrix factors through a fixed 256×112 integer matrix of rank 76, so its rank over $\mathbb{F}_q$ is at most 76 for every module rank k , and exactly 76 for generic seeds once $k \geq 7$, whereas a uniform matrix of the same shape has rank $16k$. Recovering the noise then reduces to an LWE instance of dimension 76, which the lattice estimator places at about 2^{39} operations. Second, replacing the matrix by a uniform one does not help at the proposed parameters ($n = 512$, $q \approx 2^{32}$, centred binomial noise with $\eta = 2$, ternary secrets of weight 64): the estimator gives about 2^{51} . Third, decryption at those parameters is deterministic: the noise is bounded by 258, far below $q/4$; a failure rate of $2^{-6.4 \times 10^{15}}$ reported earlier came from a Gaussian tail evaluated outside the support of a bounded sum. All computations are reproducible from public scripts.

1 Introduction

SLWE was proposed inside a larger, physics-motivated programme that uses the sedenions $\mathbb{S}$ (the Cayley–Dickson double of the octonions) and their zero divisors. The idea was to keep the shape of Module-LWE [8, 5] while replacing the ring by $\mathbb{S}_q = \mathbb{S} \otimes \mathbb{F}_q$ and to give the public matrix a symmetry inherited from the Fano plane: each row is the orbit of a random seed under a cyclic permutation σ of order 7. Correctness was obtained with a bilinear “norm” pairing rather than the (non-associative) algebra product. The scheme was never published as secure; its documentation listed the hardness of its structured matrix as open.

This note settles the question negatively and records three facts that may be useful beyond this proposal.

1. A coordinate permutation applied to seeds produces a matrix that is linear in the seeds through a fixed “design matrix”. The rank of that design matrix bounds the rank of the public matrix for every module rank (Section 3).
2. Non-associativity of the algebra plays no role once decryption uses a bilinear form: every map the attacker needs is $\mathbb{F}_q$ -linear on $\mathbb{F}_q^{16k}$.
3. At $q \approx 2^{32}$ with noise of standard deviation about 1, LWE in dimension 512 is easy whatever the matrix (Section 4). The modulus-to-noise ratio, not the algebra, governs security.

2 The scheme

Let $\mathbb{S}_q$ be the sedenions over $\mathbb{F}_q$ with basis $e_0, \dots, e_{15}$, built by four Cayley–Dickson doublings $(a, b)(c, d) = (ac - \bar{d}b, da + b\bar{c})$. Write $L(a) \in \mathbb{F}_q^{16 \times 16}$ for left multiplication by a and $\langle u, v \rangle = \text{Re}(\bar{u}v) = \sum_t u_t v_t$ for the norm pairing, extended to $\mathbb{S}_q^k$ by summing over components. The identity $L(a)^\top = L(\bar{a})$ gives $\langle As, r \rangle = \langle s, A^H r \rangle$ with $A_{ij}^H = \overline{A_{ji}}$, which is what decryption needs.

- **KeyGen.** $s \in \mathbb{S}_q^k$ is sparse ternary (weight h_s). $A \in \mathbb{S}_q^{k \times k}$ has $A_{ij} = \sigma^j(\text{seed}_i)$ with independent uniform seeds, where σ moves the coefficient of e_t to $e_{(t \bmod 7)+1}$ for $t = 1..7$, moves that of e_{t+8} to $e_{(t \bmod 7)+9}$, and fixes e_0, e_8 . The noise e is centred binomial (or confined to the four-dimensional kernel of a zero divisor [7]). The public key is $(A, b = As + e)$.
- **Encaps.** r is sparse ternary (weight h_r), with $c_1 = A^H r + e_1$ and $c_2 = \langle b, r \rangle + e_2 + m \lfloor q/2 \rfloor$.
- **Decaps.** Round $c_2 - \langle s, c_1 \rangle = m \lfloor q/2 \rfloor + N$ with $N = \langle e, r \rangle - \langle s, e_1 \rangle + e_2$.

The proposed parameters are $k = 32$ (so $n = 16k = 512$), $q = 4\,294\,977\,961$ (the least prime $\equiv 1 \pmod{455}$ above 2^{32}), $h_s = h_r = 64$, and noise $\text{CBD}(\eta = 2)$.

3 The rank collapse

Let $M \in \mathbb{F}_q^{16k \times 16k}$ be the matrix of $s \mapsto As$, i.e. $M_{16i+u, 16j+c} = \text{coefficient } u \text{ of } \sigma^j(\text{seed}_i) e_c$.

Proposition 1. *Let $N \in \mathbb{Z}^{256 \times 112}$ have entries $N_{16m+u, 16j+c} = \text{coefficient } u \text{ of } \sigma^j(e_m) e_c$ ($0 \leq m, u, c \leq 15, 0 \leq j \leq 6$). Then $M = (S \otimes I_{16}) N_k$, where $S \in \mathbb{F}_q^{k \times 16}$ holds the seeds and N_k consists of the column blocks $j \bmod 7$ of N , $j = 0, \dots, k-1$. Hence $\text{rank } M \leq \text{rank } N$ for every k . Moreover $\text{rank}_{\mathbb{Q}} N = 76$, and $\text{rank } N = 76$ modulo 911 and modulo q .*

Proof. The product is bilinear and σ is linear, so $\sigma^j(\text{seed}_i) e_c = \sum_m (\text{seed}_i)_m \sigma^j(e_m) e_c$. That is the stated factorisation. Since $\sigma^7 = \text{id}$, column block j of N_k equals block $j \bmod 7$, so N_k has at most the columns of N . The ranks of N were computed exactly (Section 7). $\square$

The bound is attained. Table 1 lists exact ranks from two independent implementations (one built on the proposal’s reference code, one written blind from a textual specification), with three seed sets per case and a uniform control. The collapse starts at $k = 5$; from $k = 7$ on the rank is 76 for every seed set tried.

k	1	2	3	4	5	6	7	8	16	32	64
$16k$	16	32	48	64	80	96	112	128	256	512	1024
rank M (Singer)	16	32	48	64	72	74	76	76	76	76	76
rank M (uniform)	16	32	48	64	80	96	112	128	256	512	1024

Table 1: Exact ranks over $\mathbb{F}_{911}$ (the values for $k \in \{8, 16, 32, 64\}$ are identical over $\mathbb{F}_q$ with $q = 4\,294\,977\,961$, for three seed sets each, in both implementations).

Remark 1. σ is not an automorphism of the sedenion product, so the “symmetry” of A is a relabelling of coordinates. The argument above uses only linearity.

Corollary 1. *Let V be the column space of M ($\dim V = \rho \leq 76$). Then $e \in b + V$. Choose ρ coordinates I on which a basis of V is invertible. Then $e_{\bar{I}} = -C e_I + (b_{\bar{I}} + C b_I)$ for a public C , an LWE instance with secret e_I of dimension ρ and $16k - \rho$ samples. The same holds for e_1 and the column space of M^H . Given e and e_1 , any r' with $M^H r' = c_1 - e_1$ gives $c_2 - \langle b - e, r' \rangle = m \lfloor q/2 \rfloor + \langle e, r \rangle + e_2$, which reveals m .*

For $\rho = 76$, $q = 4\,294\,977\,961$, CBD(2) secret and noise and $m = 436$ samples, the lattice estimator [2, 3] reports $2^{38.6}$ operations (uSVP and BDD). Its block size is at the floor of its model ($\beta = 40$), so reduction at the scale of LLL already suffices. Distinguishing (A, b) from uniform is easier still: it needs only Gaussian elimination, because A itself is rank-deficient.

4 A uniform matrix does not repair the parameters

A later revision of the proposal replaced the Singer matrix by a uniform one. Table 2 shows the estimator’s default attack set at the proposed parameters with a uniform matrix. The cheapest attack costs about 2^{51} . An independent implementation of the primal “2016 estimate” [1] (smallest β with $\sigma\sqrt{\beta} \leq \delta_\beta^{2\beta-d-1} \text{Vol}^{1/d}$, the secret rescaled to the noise width) gives $\beta = 72$, $d = 937$, a core-SVP cost of $2^{0.292\beta} = 2^{21}$ [4], the same as the estimator’s own rough uSVP figure. The cause is the modulus: with $q/\sigma \approx 2^{32}$, a block size near 70 already exposes the short vector in dimension about 940.

q	cheapest attack ($\log_2 \text{rop}$)	uSVP ($\log_2 \text{rop}, \beta$)
4 294 977 961 (proposal)	51.0 (BDD)	52.1, $\beta = 72$
2731	114.4 (dual hybrid)	122.5, $\beta = 330$
911	122.2 (BKW); 127.4 without BKW	136.9, $\beta = 383$

Table 2: Lattice estimator [3], commit 53da598, default attacks and cost model [6], at $n = 512$, $m = 512$, ternary secret of weight 64 and CBD(2) noise, with a uniform public matrix. The columns $q = 911$ and $q = 2731$ (the two smallest primes $\equiv 1 \pmod{455}$) are for orientation only.

Even the smallest admissible prime falls short of 2^{128} at $n = 512$ with these distributions, so a secure instance would need a larger dimension or wider noise. We have not searched that space.

5 Correctness is deterministic

Each coordinate of e , e_1 and e_2 lies in $[-2, 2]$. The confined zero-divisor noise has the same bound, since the kernel bases have entries in $\{0, \pm 1\}$ and the noise is $2 \sum_j \alpha_j k_j$ with $\alpha_j \in \{-1, 0, 1\}$. So

$$|N| \leq 2h_r + 2h_s + 2 = 258 < \lfloor q/4 \rfloor \approx 1.07 \times 10^9,$$

and decryption never fails at the proposed parameters. It never fails for any $q > 1032$. For CBD noise the distribution is in fact exact: given r and s , N is a sum of $2(h_r + h_s + 1)$ independent symmetric binomial draws, i.e. $N \sim \text{CBD}(258)$. At $q = 911$ this gives $\Pr[|N| > 227] = 2^{-353.5}$ exactly.

An earlier analysis reported $\log_2 \text{DFR} \approx -6.4 \times 10^{15}$. That number is the Gaussian tail at $z = q/(4\sigma_N) \approx 9.5 \times 10^7$ with $\sigma_N \approx 11.4$. It evaluates the tail about 4×10^6 times beyond the largest value N can take. The correct statement is $\text{DFR} = 0$.

6 Conclusion

The Singer-orbit public matrix makes SLWE insecure at every module rank. A uniform matrix does not rescue the proposed parameters, which reach about 2^{51} . The scheme’s correctness was never in question. The general lesson is elementary but easy to miss. A structured matrix built by acting on random seeds is the image of a fixed design matrix, and that matrix’s rank should be the first thing checked.

7 Reproducibility

All scripts, outputs and checksums are in the public repository `gifgaf0/gifgaf0.github.io` at commit

`b09929ada97cc3c5bfc0f5f327518cd82e342fea`

in the directory `audit_followup/phase_c/C1/`:

- the two rank implementations: `leg1/`, and `leg2/`, written blind from the pre-registration `C1_PREREG.md`;
- their comparison and the computation of rank N : `c1_compare.py`;
- the estimator runs: `estimator/`, using `passagemath 10.8.12` and lattice-estimator commit `53da5982597709ba0fdf94ea37a84d822310fd84`;
- the core-SVP cross-check: `c1_coresvp_crosscheck.py`;
- the noise bounds and exact tails: `c1_checks.py`.

Exact ranks were computed with `python-flint`.

Acknowledgements. AI agents (Anthropic’s Claude) drafted this note and wrote and ran all of its computations, including both independent rank implementations; the author directed the work, reviewed the text, and takes responsibility for it.

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